

Scientific Machine Learning: Blending Science with Data

Homework Days 4+5

- a) Generate fictitious data from the function

$$y(x) = \sin(x) + \epsilon$$

where ϵ is a small amount of random noise. Use the domain $x \in [0, 2\pi]$.

- b) Plot this data set.

- c) Use the model below

$$f_{model}(x) = a_0 + a_1x + a_2x^2 + a_3x^3$$

to fit the data you generated in a). Use the standard gradient descent algorithm to find the coefficients. Plot $f_{model}(x)$ with your fictitious data. Do you get a good fit? Please feel free to use the optimisation libraries in Julia taught in lectures.

- d) Try different optimisation techniques (Momentum, Adam etc) with different hyperparameters. Can you get better models?